

SHYAM PATHAK

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Profile Summary

Final-year Computer Science undergraduate with a strong foundation in machine learning, deep learning, and NLP. Proven ability to develop and deploy scalable AI/ML models, demonstrated through real-world projects like a BERT-based sentiment analysis system. Skilled in Python, model optimization, and API development for production-ready pipelines. Adaptable and analytical, with a readiness to learn and apply GenAI technologies such as LLMs, RAG pipelines, and prompt engineering to solve impactful problems

Technical Skills

Python, SQL, Java (Basic), C++ | Machine Learning: Scikit-learn, Pandas, NumPy, Feature Engineering, Model Evaluation (Precision, Recall, F1), Hyperparameter Tuning | NLP & GenAI: Text Classification, Embeddings, Prompt Engineering, RAG, LangChain, Hugging Face Transformers | Deep Learning: TensorFlow | Computer Vision & OCR: YOLO, OpenCV, Tesseract OCR, PaddleOCR | Deployment & Tools: FastAPI, Flask, MLflow, Git | Core: OOP, DSA, SDLC, Agile | Soft Skills: Analytical, Problem Solving, Communication

Experience

AI Engineering Intern

Sep 2025 – Mar 2026

[Link1](#) | [Link2](#)

In-office, Paid, India

- Engineered end-to-end Generative AI pipelines for legal document analysis using RAG, enabling accurate, context-aware insights.
- Built and optimized LLM-based agents and chat systems using LangChain/LangGraph, improving response relevance and Quality
- Developed a computer vision-based pipe take-off system using YOLO and OCR (PaddleOCR/Tesseract) to detect, extract, and quantify pipeline components from engineering drawings.
- Designed scalable document processing pipelines (chunking, embeddings, retrieval) and contributed to deploying production-ready AI systems with modular, robust architecture.

Projects

FraudX | [Link](#)

Mar 2025 – May 2025

- Built **FraudX**, a fraud detection system trained on a **6.3 million+ transaction dataset** from Kaggle, achieving **95.2% accuracy**, **94.7% precision**, and **96.1% recall** on fraud detection.
- Preprocessed and engineered features across **10+ variables**, applied **Label Encoding**, **StandardScaler**, and handled class imbalance using **SMOTE**, improving model F1-score by **18%**.

Sentiment Analysis for Movie Reviews | [Link](#)

Jan 2025 – Mar 2025

- Created a BERT-based sentiment analysis model with **88% accuracy**.
- Adopted NLP techniques (**LSTM**, **lemmatization**, **embeddings**) and deployed via **Flask API** for real time interaction achieved **88% accuracy**.
- Speeding up **readiness by 25%** and boosting **data accuracy by 15%**.

CardioSense | [Link](#)

Sep 2024 – Nov 2024

- Built a heart disease prediction model using **8 machine learning algorithms**, achieving the highest accuracy
- of **95.08% with Random Forest** on test data split (80:20).
- Analyzed a dataset of **303 patient records across 14 clinical features** including age, cholesterol, ECG results, and heart rate to identify key risk indicators.
- Conducted model comparison: **Logistic Regression (85.25%)**, **XGBoost (85.25%)**, **SVM (81.97%)**, and **Neural Network (80.33%)**, validating using **accuracy, correlation, and bar plots**.

Certifications

[SupervisedML: Regression & Classification – \]](#)

[LearningDeeplearnig.AI\)](#)

[Neural Networks and CNN Essential Training](#)

[Python for Data Science –NPTEL, Govt. of India](#)

[What Is Generative AI – LinkedIn](#)

[What Is Generative AI – LinkedIn Learning](#)

[Data Analytic– One Roadmap](#)

[ExploreML -- Springboard \(Infosys\)](#)

Leadership and Extra-Curricular Activities

- Secretary, BitWise Club**: Organized coding events, mentored juniors, and supported technical troubleshooting in club activities with **50+** students.
- Member, Google Developer Student Club**: Participated in collaborative hackathons, projects, **8+** workshops.
- Volunteered in tech-for-good drives, supporting **200+** families with efficient donation tracking systems.

Education

Bachelor of Technology CSE
Higher Secondary Education (CBSE)
Senior Secondary Education (ICSE).

Jaypee University of Engineering and Technology
K R Mangalam World School
City Montessori School

Guna – Oct 2021 to June 2025
Gurugram – 82% 2021
Lucknow – 92% 2019